## Executive Summary

- PRAMAAN-AI is a secure AI decision-support layer betwe portal and the evaluator.
- Five engines parse, prove, match, verify and see; a versio Graph evaluates machine-readable norms.
- Every claim retains source page, region and confidence. T 0–100 routes GREEN, AMBER or RED.
- AI is advisory only; an authorised human official records t

## Problem Analysis & Traceability

- Manual inspection creates late deficiencies, inconsistent s
- preventable EVC visits.
- R1 PARSE → completeness and expiry; R2 PROVE → fore
- inspection; R3 MATCH → historical and faculty graph.
- R4 VERIFY → issuer/source receipts; R5 SEE → GPS, cou
- reuse; R6 → evidence report.
- R7 queue → EVC prioritisation; R8 → hash-chain audit; R9
- decision guardrail.

## Solution Overview

- APH-as-Code: clause, applicability, evidence, threshold, s handbook edition are configuration
- Claim Graph: typed claims are the unit of verification, not document blobs
- Six steps: ingest → classify/extract → build claims → run engines → score/triage → human decision
- Configuration changes are versioned and diff-reviewed be

## The Five Verification Engines

- **PARSE: PDF text/layout seam, checklist, expiry, prescribe arithmetic checks.**
- **PROVE: ELA, metadata/font signals and copy/splice inspe an inspection request.**
- **MATCH: year-on-year deltas and salted faculty identifiers unexplained changes and ghost faculty.**
- **VERIFY: source-channel strength is ordered DigiLocker/A and registry; unavailable is manual queue.**
- **SEE: geofence, photo reuse and future object-count seam focused EVC agenda.**

## Trust Score & Triage

- Weights: completeness 20%, authenticity 25%, consistenc
20%, infrastructure 15%.
- Hard gates: unresolved HIGH caps 58; confirmed mandato
caps 40.
- GREEN ≥75 without HIGH; AMBER 50–74 or capped; RED
breach.
- Synthetic demonstration: 52/100 AMBER; authenticity coll
because of one suspect document.

## System Architecture

- MVP: FastAPI, SQLite, PyMuPDF/Pillow seams, reportlab s an offline single-page dashboard.
- Production posture: PostgreSQL, object storage, workers pool, RBAC/SSO and AICTE VPC deployment.
- Stable contracts let heavier OCR, LayoutLM, YOLO, issue PKCS#7 validation replace MVP adapters.
- Evidence and audit records are persisted independently fr

## Reviewer Experience & Workflows

- Evaluator opens the risk-ranked queue, selects a submiss
- each flag to source document/page/region.
- Accepted/overridden flags require an audit event; overrid
- reason.
- Senior Official is the only role permitted to Approve, Rejec
- cure.
- RED creates a ready-made EVC agenda: inspect the exact
- document or issuer claim first.

## Prototype Implementation & Results

- Standalone engine and live app are included in this repos
values are synthetic.
- Reproducible output: clean/tampered image ELA, 52-point
flags and a valid seven-entry chain.
- Live upload path: multipart evidence → extraction → five-
queue → decision → audit.
- Prototype output is real code output; target impact figures
as measured.

## Security, Privacy & Governance

- Role matrix: Evaluator reviews, Senior Official decides, Ac
Auditor verifies.
- Salted faculty graph keys avoid exposing raw identity in a
minimised and deployment can be on-prem.
- SHA-256 previous-hash links make changes detectable; c
is a reviewer action.
- Random GREEN re-checks, per-engine shadow metrics ar
routing address automation bias.

## Feasibility, Roadmap & Delivery

- Phase 0: offline pilot; every flag must be evidence-linked.
- shadow mode beside manual scrutiny.
- Phase 2: assisted live triage and deficiency view. Phase 3
- types and next APH rule pack.
- Poor scans → OCR ensemble and human routing; unavail
- manual queue; sophisticated forgery → defence in depth.
- Kill criterion: underperforming engines stay advisory-only

## Impact, Benefits & Measurement

- Design targets: ~80% less pre-scrutiny effort, feedback in
desk-clearable.
- Measure minutes/application, time-to-first-deficiency, flag
EVC precision and override rate in shadow mode.
- Institutions gain predictable cure loops; evaluators spend
judgement; AICTE gains fraud-pattern intelligence.
- Fewer repeat visits and paper handling support environm

## Risks, Limitations & Ethics

- A flag is not a finding of forgery. Human officials retain au applicants retain a cure path.
- Regional-language scans, false positives and adversarial stratified validation, not one average accuracy.
- MVP interfaces OCR, YOLO, live issuer APIs and producti does not pretend those integrations are live.
- Data governance, retention, access logging and issuer ag delivery gates.

## Research & References

- AICTE Approval Process Handbook 2024–25 to 2026–27; DigiLocker Issuer/Requester API specifications and API S (NeGD/MeitY).
- LayoutLMv3 (ACM Multimedia 2022); PaddleOCR; ELA and literature; Ultralytics YOLO.
- NIST SP 800-92; OWASP logging/RBAC guidance; PDF CM validation practice.

## Appendices · Rule Extract · File Guide · Glossary

- APH-DOC-001: fire safety certificate · EoA/New · unexpire
- APH-EXT-001: issuer/source receipt · EoA/New · verified c HIGH.
- Run: pramaan-app/run.sh for the application; python3 pramaan-ai/engine/pramaan_engine.py for the offline engi
- APH = Approval Process Handbook · DSC = Digital Signat ELA = Error Level Analysis · EVC = Expert Visiting Commi